

Query 1:

```
CREATE EXTENSION postgis;
```

Query 2:

```
CREATE TABLE locations (  
  id INT PRIMARY KEY,  
  name VARCHAR(50),  
  geo_location GEOMETRY(Point,4326)  
);
```

Query 3:

```
INSERT INTO locations (id, name, geo_location) VALUES  
-- Home  
(1, 'Home', ST_SetSRID(ST_MakePoint(-118.28774732502247, 34.03248127197735), 4326)),  
  
-- Libraries  
(2, 'Leavey Lib', ST_SetSRID(ST_MakePoint(-118.283032, 34.021632), 4326)),  
(3, 'Doheny Lib', ST_SetSRID(ST_MakePoint(-118.2841969324821, 34.02027781641586),  
4326)),  
(4, 'Helen Lib', ST_SetSRID(ST_MakePoint(-118.287774, 34.019377), 4326)),  
  
-- Eateries  
(5, 'Burger Crush', ST_SetSRID(ST_MakePoint(-118.286437, 34.020127), 4326)),  
(6, 'Rock Reillys', ST_SetSRID(ST_MakePoint(-118.28411570742323, 34.02439993118859),  
4326)),  
(7, 'Dulce', ST_SetSRID(ST_MakePoint(-118.28534697081211, 34.02549445524561), 4326)),  
  
-- Fountains  
(8, 'RTCC Fountain', ST_SetSRID(ST_MakePoint(-118.286128, 34.020421), 4326)),  
(9, 'Leavey Fountain', ST_SetSRID(ST_MakePoint(-118.283239, 34.022229), 4326)),  
(10, 'Trojan Fountain', ST_SetSRID(ST_MakePoint(-118.28530389010646,  
34.02033916151937), 4326)),  
  
-- Department Buildings  
(11, 'Cinematic Arts', ST_SetSRID(ST_MakePoint(-118.28778036459647,  
34.02396245145261), 4326)),  
(12, 'Viterbi', ST_SetSRID(ST_MakePoint(-118.289495, 34.020647), 4326)),  
(13, 'Dornslife', ST_SetSRID(ST_MakePoint(-118.287632, 34.019714), 4326));
```

Query 4:

```
SELECT ST_AsKML(ST_ConvexHull(ST_Collect(geo_location))) FROM locations;
```

Output:

#	st_askml
1	<Polygon><outerBoundaryIs><LinearRing><coordinates>-118.287774,34.019377 -118.289495,34.020647 -118.28774732502247,34.03248127197735 -118.28411570742323,34.02439993118859 -118.283239,34.022229 -118.283032,34.021632 -118.2841969324821,34.02027781641586 -118.287774,34.019377</coordinates></LinearRing></outerBoundaryIs></Polygon>

Query 5:

```
SELECT neighbor.name,ST_AsText(neighbor.geo_location) AS coordinates FROM
  locations AS neighbor, locations AS home
WHERE home.name = 'Home' AND neighbor.name != 'Home' ORDER BY
  home.geo_location <-> neighbor.geo_location LIMIT 4;
```

Output:

#	name	coordinates
1	Dulce	POINT(-118.28534697081211 34.02549445524561)
2	Cinematic Arts	POINT(-118.28778036459647 34.02396245145261)
3	Rock Reillys	POINT(-118.28411570742323 34.02439993118859)
4	Leavey Fountain	POINT(-118.283239 34.022229)